

MASOOD NAZARI

Clinical Data Coordinator | Data Engineer | Healthcare & FinTech

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PERSONAL PROFILE

I am a Clinical Data Coordinator and Data Engineer with an MSc Distinction and five years of experience building data systems that actually get used in high-stakes environments. My current role sits at the intersection of clinical research and software engineering: I coordinate and automate the data infrastructure for live NHS oncology trials at the University of Southampton, where the data has to be right because regulatory compliance and patient safety depend on it.

The things I am good at are designing ETL pipelines that hold under real-world conditions, building data quality frameworks from scratch, and making complex data problems legible to the people who need to act on them. I have reduced ingestion error rates to **under 0.1%**, cut manual workloads by **70%+**, and built audit systems that satisfy GDPR, GxP, and formal regulatory review. My core stack is Python, SQL, FastAPI, Docker, and Azure, with experience across FinTech, healthcare, and geospatial analytics.

TECHNICAL SKILLS

- **Data Engineering:** ETL/ELT Pipelines, Data Warehouse Design, Schema Validation, Data Lineage, Data Quality Frameworks, Automated QA, Data Reconciliation, Pydantic Schema Enforcement.
- **Languages:** Python (Pandas, GeoPandas, Shapely, FastAPI), SQL (PostgreSQL/T-SQL), R, Bash, SAS.
- **Cloud and Infrastructure:** Azure (ACI), GCP, Docker, Terraform (IaC), Google Earth Engine, API Integration.
- **ML and Analytics:** Scikit-learn, XGBoost, CatBoost, Prophet, SHAP/LIME, Neural Networks (BERT, SpaCy).
- **BI and Visualisation:** Streamlit, Power BI, Tableau, R Shiny, Plotly, Excel (Advanced).
- **Compliance:** GDPR, GxP (ICH/FDA), Regulatory Audit Trail Design, Metadata Governance, FCA/Basel III (Transferable).

PROFESSIONAL EXPERIENCE

Clinical Data Coordinator

Southampton Clinical Trials Unit (SCTU) | University of Southampton

Sep 2024 – Present

Southampton, UK

I work on the data engineering, automation, and reporting infrastructure for the Cancer Vaccine Launch Pad programme: three live NHS oncology trials (BNT122-01, BNT113-01, SCOPE) running across 70+ external clinical sites.

- **Pipeline design and delivery:** The unit was reconciling adverse event data from 70+ external sites manually, with no schema enforcement and recurring data quality failures. I designed and deployed an event-driven RESTful pipeline in Python and FastAPI with Pydantic validation at ingestion. Errors dropped to under **0.1%**. Anomalies are now flagged automatically, before they reach the analysts.
- **Real-time reporting platform:** Trial Managers were getting data in weekly batches, which meant problems were often identified too late to fix without disrupting trial timelines. I built a containerised reporting layer (Streamlit, Docker) integrated directly with the pipeline. They can now spot discrepancies **6x faster** than before, and act on them the same day.
- **Audit and compliance infrastructure:** Clinical trial data has to be traceable. Every transformation, every ingestion, every correction needs a record that can survive regulatory scrutiny. I implemented immutable audit-log pipelines satisfying both GDPR and GxP requirements, which cut QA retrieval time by **80%** and gave the unit a compliance framework they did not have before.
- **Security and infrastructure review:** I carried out an architectural review of the third-party integration tools across the data estate and found OAuth authentication vulnerabilities that were exposing data access across all external sites. I wrote the remediation roadmap and worked it through the governance process until it was fixed.
- **Statistical automation:** Automated the recurring SAS and R reporting workflows for two studies, removing the manual analyst time that had been absorbed into every reporting cycle.

Data Engineer (Internship)

SC Solutions | Environmental Analytics

Jun 2024 – Sep 2024

Remote/Hybrid

Built geospatial data pipelines for satellite-based biomass prediction across large-scale land areas.

- **Geospatial ETL:** Designed automated pipelines using Python (GeoPandas, Shapely) and Google Earth Engine to ingest and transform multi-terabyte satellite imagery from NASA GEDI and Sentinel SAR/Optical sources. The data came from different sensors, in different formats, at different resolutions. Making it usable took both engineering and domain understanding.
- **Data quality and validation:** Built topology and geometry validation scripts to catch and resolve errors in the satellite data before processing. Without this step, the noise in the raw imagery would have propagated into the models.
- **Performance work:** Restructured the preprocessing pipeline using parallelised feature extraction, reducing latency by **80%** and compressing the full cycle from days to hours. Deployed ML models across **10,000+ hectares** with SHAP-based explainability outputs for carbon stock estimation.

Senior Data Analyst / Data Engineer

Parsian E-Commerce Co. | FinTech

Oct 2019 – Sep 2023

Tehran

Data engineering and analytics lead at a fast-growing Iranian e-commerce and payments platform.

- **Fraud detection:** Built ML-based transaction anomaly detection pipelines from scratch. Fraud capture rates improved by **26%**. In a high-volume payments environment, that is a material reduction in financial exposure.
- **Reconciliation automation:** Financial reconciliation was being done manually, every month. I replaced it with SQL and Python-based pipelines that cut manual handling time by **43%** and took two days off the month-end close cycle.
- **Business intelligence:** Built the executive dashboard layer on top of structured data warehouse queries, giving leadership real-time visibility into revenue and cost drivers and improving decision-making efficiency by approximately **27%**.

Data Analyst

MaximTaxi | Private Transport Platform

Jun 2018 – Sep 2019

Tehran

- Used SQL (PostgreSQL) to identify financial discrepancies in transactional data, improving supplier payment accuracy by **23%**. Built automated compliance reporting queries to replace manual Excel processes.

TECHNICAL PROJECTS

Fraud Detection Pipeline API | Open Source

2024

- **Stack:** Python, FastAPI (Async), NetworkX, Docker, Pydantic, Pandas.
- A high-concurrency async backend processing **5 million+ financial transactions** at sub-100ms latency. Pydantic schema enforcement handles type-safe ingestion at volume. Graph theory algorithms (NetworkX) detect circular fraud patterns. Explainable AI outputs (XAI) provide audit-ready rejection transparency for regulatory compliance.

NIHR Research Intelligence Dashboard | Open Source

2024

- **Stack:** Python, Streamlit, Scikit-learn, Pandas, Plotly.
- Built a portfolio intelligence platform governing **£171.6M** in research funding. Fuzzy Matching deduplication pipeline detected a 5.6% duplicate award rate and enforced NIHR data governance rules, raising dataset completeness to **73.6%**. Automated parliamentary-facing reporting workflows, delivering verified **122:1 ROI** in saved analyst time.

BioVision Analytics Hub (Geospatial ETL) | Open Source

2024

- **Stack:** Python, GeoPandas, LightGBM, XGBoost, Sentinel-1/2 APIs, Streamlit, Mapbox.
- Automated ETL pipeline for multi-terabyte geospatial datasets with parallelised feature extraction and geometry validation. Achieved **0.89 R²** accuracy for forest ecosystem monitoring. Formed the basis of an MSc Distinction dissertation on AI-driven geospatial forecasting.

InsurePrice: Actuarial Risk Engine | Open Source

2024

- **Stack:** Python, CatBoost, SHAP, Terraform (IaC), Azure Container Instances, FastAPI.
- Deployed a containerised ML pricing engine on Azure using Terraform. Full dev/prod parity via Docker. Monte Carlo risk simulation across 10,000 commute scenarios. SHAP integration for audit-ready explainability outputs.

Bank Marketing Analytics Platform | Open Source

2024

- **Stack:** Python, React, Flask (REST API), Streamlit, Scikit-learn (SMOTE).
- Dual-interface deployment: Streamlit for rapid prototyping, decoupled Flask/React REST API for production across 7 endpoints. SMOTE class imbalance handling raised minority class recall to **60%+**.

EDUCATION

MSc Data & Decision Analytics — Distinction

University of Southampton, UK

Sep 2023 – Sep 2024

- **Dissertation:** “AI-driven geospatial forecasting” — Distinction. Deployed as the BioVision Analytics Hub.
- **Core modules:** Advanced Machine Learning, Data Mining, Stochastic Processes, Operational Research.

BSc Materials Engineering — Distinction, Top 5% of cohort

Semnan University

Sep 2010 – Sep 2014

- Specialised in computational modelling and simulation.